



PANAMA

About labs in this class

The labs in this class will have general instructions, and many things need to be figured out by the students. I will be answering any specific questions the students may have without completely giving away the key to the puzzle.

Answer the questions and record your measurements in your lab notebook, and then submit the notebook at the end of the activity.

About this lab

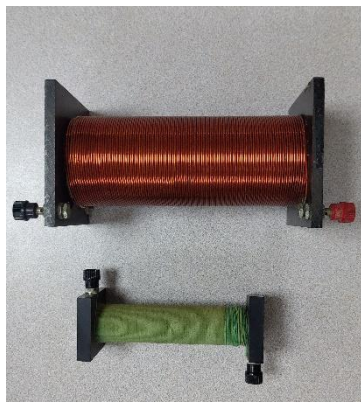
In this lab, you are provided with all the tools necessary to light a lightbulb wirelessly. You have two solenoids, one, connected to a power supply, will produce a magnetic flux change in the other one (see figure 1). You are provided with two power supplies: the Universal Interface, and an old school power supply that can produce DC and AC Voltages.

Activity 1. Light the light bulb via electromagnetic induction using DC.

Question 1. What is the current through the lightbulb? Do a theoretical estimation (you will have to estimate the magnetic field flux change through the smaller solenoid) and then measure it with the tools provided.

Question 2. What is the circuit diagram of this circuit you have created?

Activity 2. Using an AC power supply, light the light bulb. Make it so that it remains on, without any further human interference other than turning on the power. Draw a diagram of the situation.



Question 3: Estimate the current through the second solenoid and then measure it. Record the value. Show the work for your theoretical estimate.

Figure 1: The solenoids in question.

Manual to Lab 7: PHY2049C.

Florida State University

Electromagnetic Induction

